Pipeline ML On-Chain v3
Token UNI (Uniswap) | LONG/SHORT/FLAT

Generado: 19/04/2026 21:26

Frecuencias: 4h, daily (se descarta 1h)

Variante A: XGBoost umbral fijo | B: XGBoost F&G dinamico | C: LSTM fijo

Capital: $10,000 | Trade size: $500 | Fee: 0.2% round trip

Mejor combinacion: B-4h | Net P&L: $+804.08

1. Cambios respecto a V2

V3 introduce cambios exclusivamente en la capa de decision y simulacion.
Los modelos XGBoost y LSTM se reutilizan de V2 sin reentrenamiento.

Cambios principales:

1. Se descarta la frecuencia 1h completamente.
2. Se agrega estado FLAT: tres estados posibles LONG / SHORT / FLAT.
3. Umbral dinamico basado en Fear & Greed para XGBoost (variante B).
4. Umbral fijo asimetrico para LSTM (variante C).
5. Capital inicial $10,000 USD con trade size fijo de $500.
6. Reporte financiero detallado en USD con tracking por trade.

Logica de umbrales:

```
LONG  si proba > umbral_long
SHORT si proba < umbral_short
FLAT  si umbral_short <= proba <= umbral_long
```

```
Variante A (XGB fijo):      long=0.60  short=0.40
Variante B (XGB F&G):
  Extreme Fear (0-25):      long=0.72  short=0.32
  Fear (25-45):             long=0.65  short=0.38
  Neutral (45-55):          long=0.60  short=0.40
  Greed (55-75):            long=0.55  short=0.45
  Extreme Greed (75-100):   long=0.52  short=0.48
Variante C (LSTM fijo):     long=0.58  short=0.42
```

Contabilidad USD:

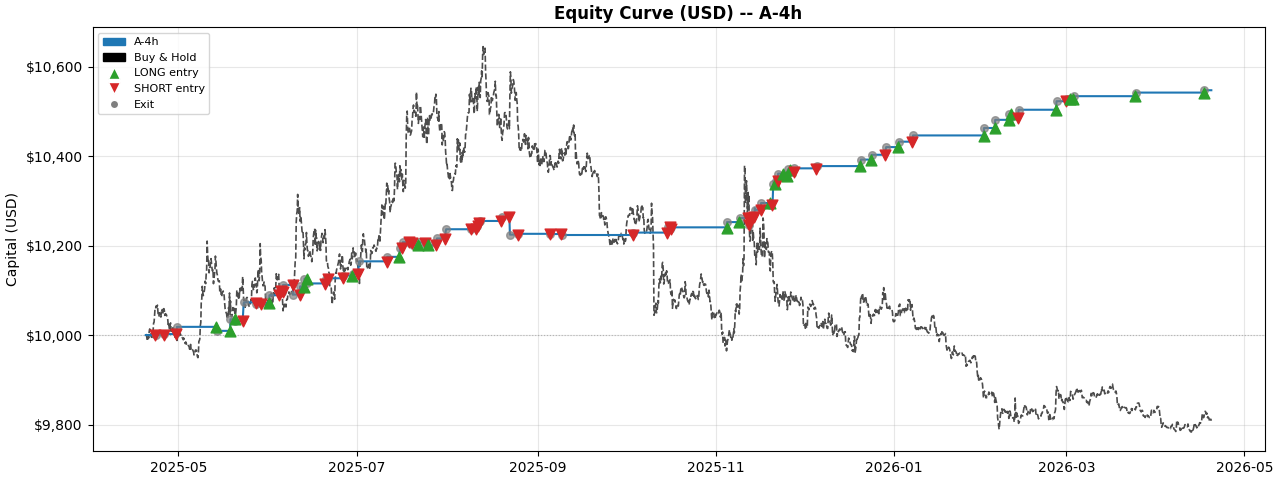
- Capital base $10,000 en USD. Trade size $500 fijo.
- Fee 0.1% por entrada + 0.1% por salida = 0.2% round trip.
- Solo una posicion abierta a la vez.
- Al cambiar de LONG a SHORT: cierra LONG (fee salida) + abre SHORT (fee entrada).

2. Resultados -- A-4h

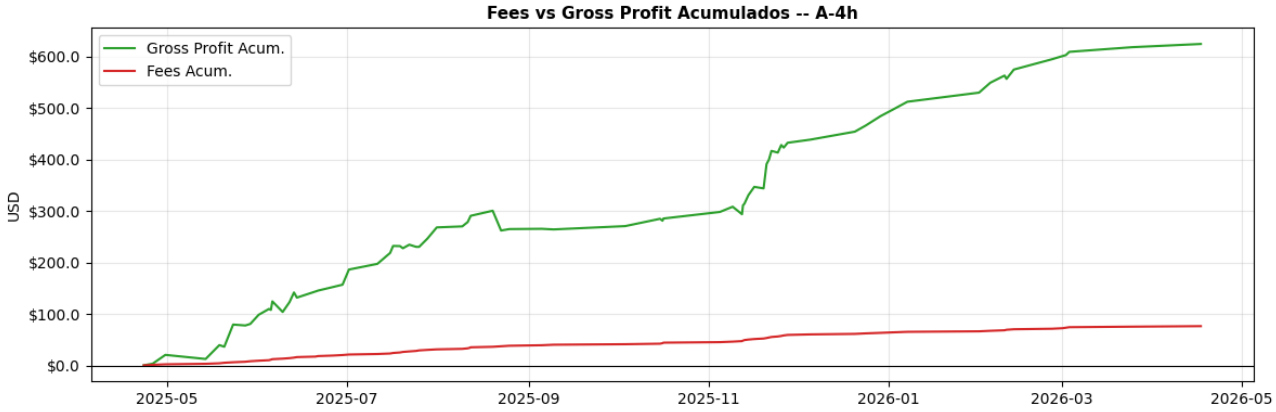
2.1 Tabla de P&L en USD

```
Capital inicial:        $   10,000.00
Capital final:          $   10,547.09
Profit/Loss neto:       $     +547.09  (+5.47%)
---------------------------------------------
Total trades:                      77
Trades LONG:                       29
Trades SHORT:                      48
Periodos en FLAT:           2104  (96.1%)
---------------------------------------------
Trades ganadores:             58  (75.3%)
Trades perdedores:                 19
---------------------------------------------
Gross profit:           $      753.91
Total fees pagadas:     $       77.00
Net profit:             $     +547.09
---------------------------------------------
Mejor trade:            $      +45.96
Peor trade:             $      -39.38
Racha ganadora max:           10 trades
Racha perdedora max:           2 trades
---------------------------------------------
Sharpe ratio:                   3.889
Max drawdown:              -0.40%  ($-40.74)
Dias en test:                     364
```

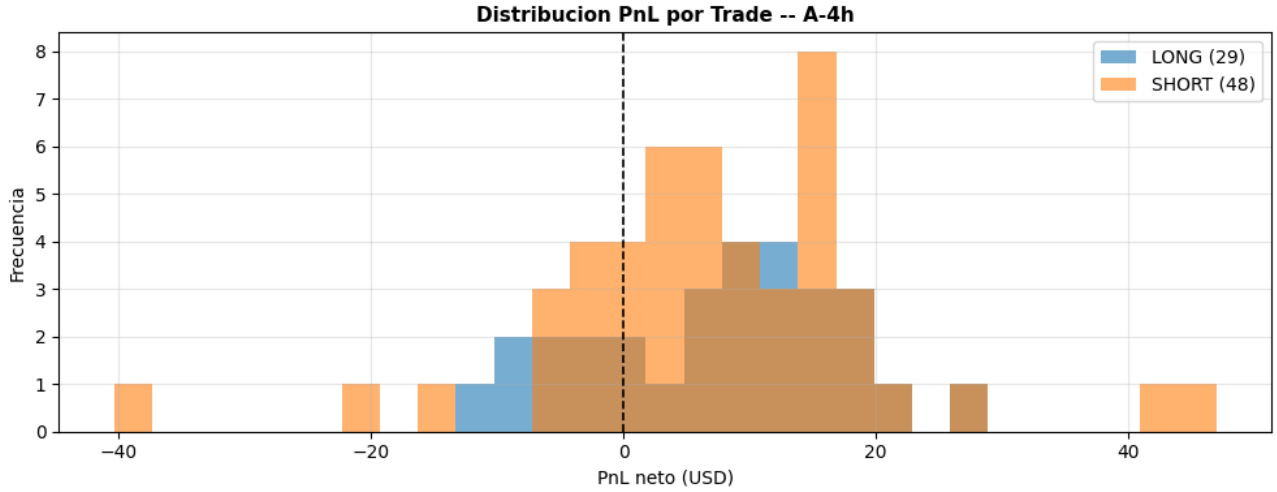
2.2 Equity Curve en USD



2.3 Fees Acumuladas vs Profit Bruto



2.4 Distribucion PnL por Trade



2.5 Log de Trades (primeros 20)

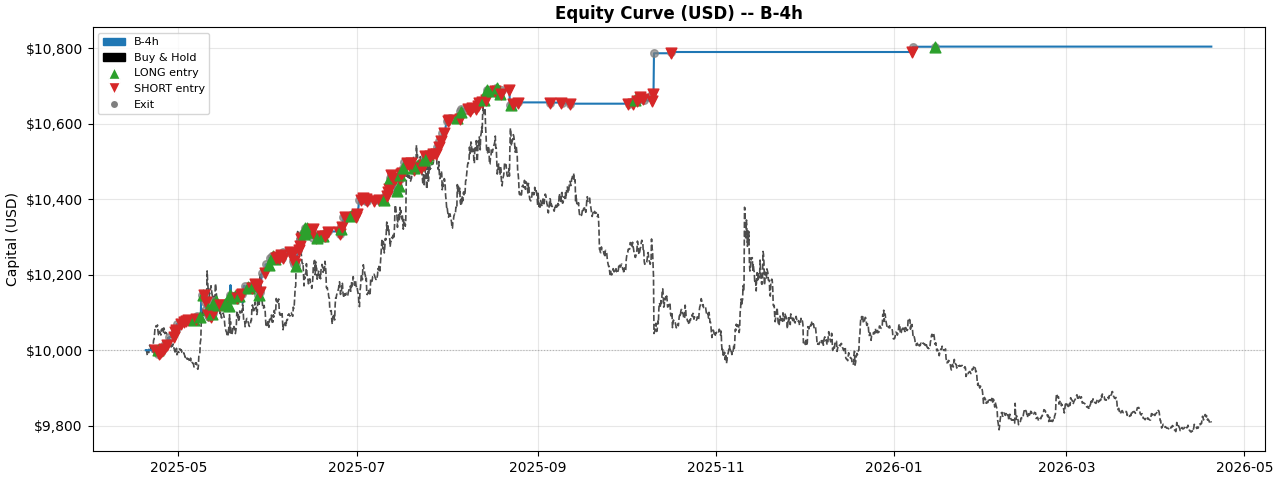
Entrada	Salida	Dir	P.entry	P.exit	PnL neto	Fee	Capital
2025-04-23 00:00	2025-04-23 04:00	SHORT	$5.9540	$5.9410	$+0.09	$1.00	$10,000.09
2025-04-26 00:00	2025-04-26 04:00	SHORT	$5.9500	$5.9130	$+2.11	$1.00	$10,002.20
2025-04-30 08:00	2025-04-30 12:00	SHORT	$5.3770	$5.1930	$+16.11	$1.00	$10,018.31
2025-05-14 00:00	2025-05-14 04:00	LONG	$6.9040	$6.7940	$-8.97	$1.00	$10,009.34
2025-05-18 16:00	2025-05-18 20:00	LONG	$5.8150	$6.1280	$+25.91	$1.00	$10,035.26
2025-05-20 08:00	2025-05-20 12:00	LONG	$5.8270	$5.7900	$-4.17	$1.00	$10,031.08
2025-05-23 00:00	2025-05-23 12:00	SHORT	$6.7110	$6.1340	$+41.99	$1.00	$10,073.07
2025-05-27 12:00	2025-05-27 16:00	SHORT	$6.5110	$6.5340	$-2.77	$1.00	$10,070.31
2025-05-29 04:00	2025-05-29 08:00	SHORT	$7.1640	$7.1210	$+2.00	$1.00	$10,072.31
2025-06-01 00:00	2025-06-01 04:00	LONG	$5.9690	$6.1800	$+16.67	$1.00	$10,088.98
2025-06-04 12:00	2025-06-04 16:00	SHORT	$6.4810	$6.3340	$+10.34	$1.00	$10,099.32
2025-06-05 04:00	2025-06-05 08:00	SHORT	$6.2580	$6.2820	$-2.92	$1.00	$10,096.40
2025-06-05 16:00	2025-06-05 20:00	SHORT	$6.1010	$5.8980	$+15.64	$1.00	$10,112.04
2025-06-09 04:00	2025-06-09 08:00	SHORT	$6.2780	$6.5360	$-21.55	$1.00	$10,090.49
2025-06-11 12:00	2025-06-11 16:00	SHORT	$8.3490	$8.0280	$+18.22	$1.00	$10,108.72
2025-06-13 00:00	2025-06-13 04:00	LONG	$7.1340	$7.3990	$+17.57	$1.00	$10,126.29
2025-06-14 00:00	2025-06-14 04:00	LONG	$7.4850	$7.3360	$-10.95	$1.00	$10,115.34
2025-06-20 08:00	2025-06-20 12:00	SHORT	$7.7140	$7.5300	$+10.93	$1.00	$10,126.26
2025-06-21 04:00	2025-06-21 08:00	SHORT	$6.9460	$6.9230	$+0.66	$1.00	$10,126.92
2025-06-26 08:00	2025-06-26 12:00	SHORT	$6.9470	$6.8490	$+6.05	$1.00	$10,132.97

2. Resultados -- B-4h

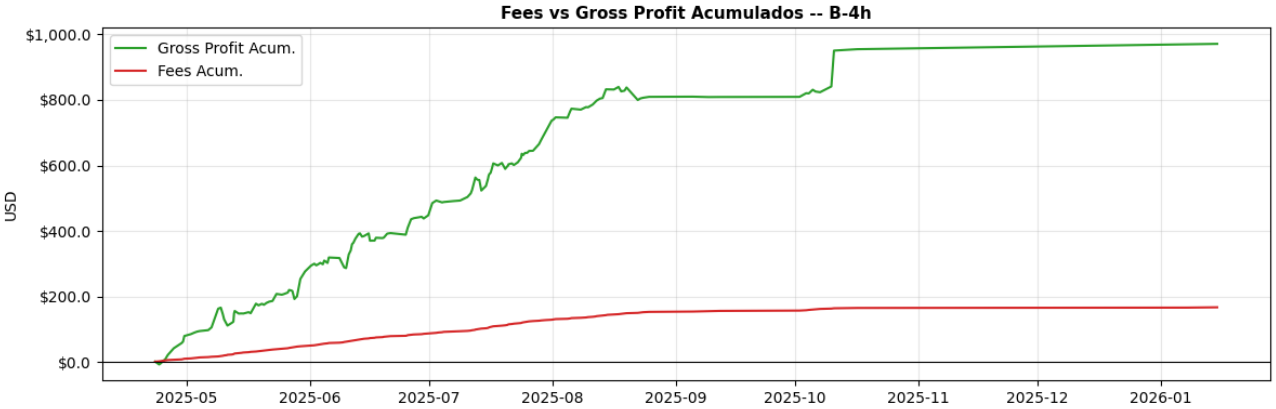
2.1 Tabla de P&L en USD

```
Capital inicial:        $   10,000.00
Capital final:          $   10,804.08
Profit/Loss neto:       $     +804.08  (+8.04%)
---------------------------------------------
Total trades:                     167
Trades LONG:                       56
Trades SHORT:                     111
Periodos en FLAT:           1959  (89.5%)
---------------------------------------------
Trades ganadores:            114  (68.3%)
Trades perdedores:                 53
---------------------------------------------
Gross profit:           $    1,323.05
Total fees pagadas:     $      167.00
Net profit:             $     +804.08
---------------------------------------------
Mejor trade:            $     +108.39
Peor trade:             $      -39.38
Racha ganadora max:           16 trades
Racha perdedora max:           4 trades
---------------------------------------------
Sharpe ratio:                   3.407
Max drawdown:              -0.66%  ($-66.95)
Dias en test:                     364
```

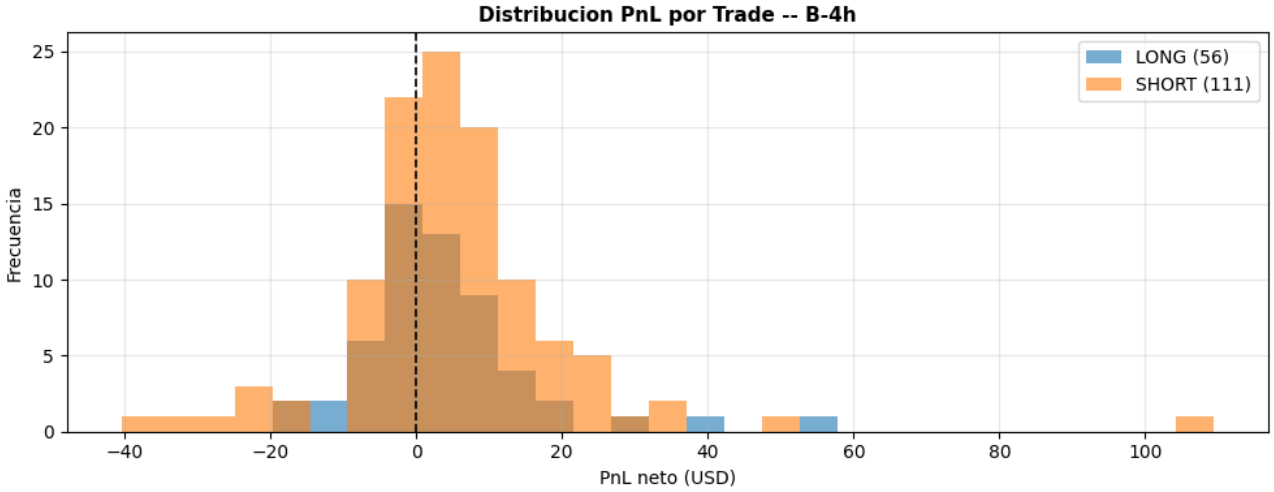
2.2 Equity Curve en USD



2.3 Fees Acumuladas vs Profit Bruto



2.4 Distribucion PnL por Trade



2.5 Log de Trades (primeros 20)

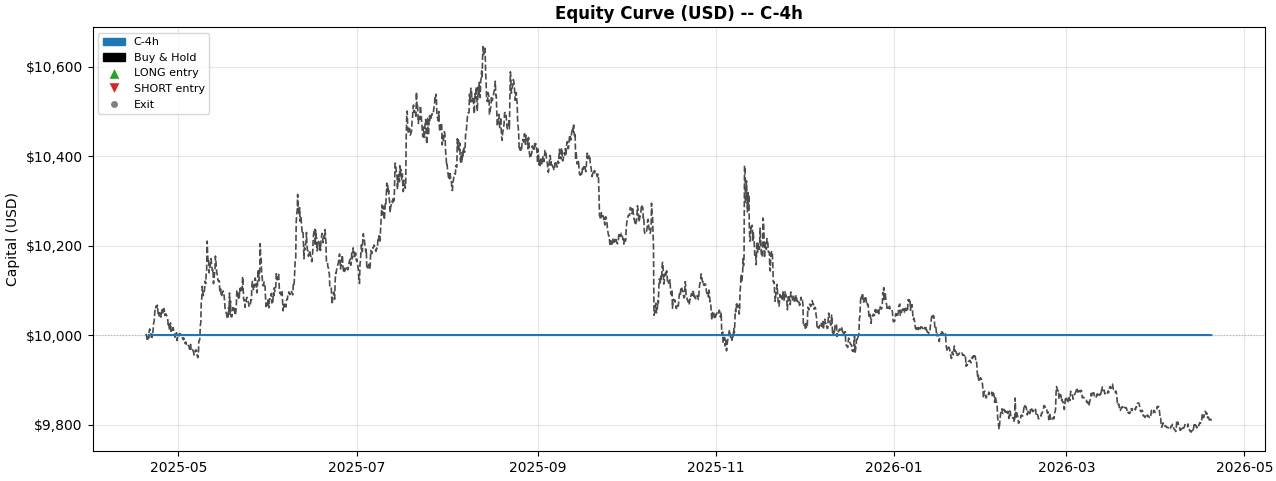
Entrada	Salida	Dir	P.entry	P.exit	PnL neto	Fee	Capital
2025-04-23 00:00	2025-04-23 04:00	SHORT	$5.9540	$5.9410	$+0.09	$1.00	$10,000.09
2025-04-24 00:00	2025-04-24 04:00	LONG	$5.8600	$5.7590	$-9.62	$1.00	$9,990.47
2025-04-24 12:00	2025-04-24 16:00	SHORT	$5.8520	$5.8000	$+3.44	$1.00	$9,993.92
2025-04-24 20:00	2025-04-25 00:00	SHORT	$5.8110	$5.7490	$+4.33	$1.00	$9,998.25
2025-04-25 12:00	2025-04-25 16:00	SHORT	$5.9300	$5.8640	$+4.56	$1.00	$10,002.82
2025-04-26 00:00	2025-04-26 08:00	SHORT	$5.9500	$5.7890	$+12.53	$1.00	$10,015.35
2025-04-27 04:00	2025-04-27 20:00	SHORT	$5.7360	$5.5000	$+19.57	$1.00	$10,034.92
2025-04-29 08:00	2025-04-29 20:00	SHORT	$5.4460	$5.2710	$+15.07	$1.00	$10,049.98
2025-04-30 00:00	2025-04-30 04:00	SHORT	$5.3350	$5.2890	$+3.31	$1.00	$10,053.30
2025-04-30 08:00	2025-04-30 12:00	SHORT	$5.3770	$5.1930	$+16.11	$1.00	$10,069.41
2025-05-02 00:00	2025-05-02 04:00	SHORT	$5.3260	$5.2640	$+4.82	$1.00	$10,074.23
2025-05-02 20:00	2025-05-03 00:00	SHORT	$5.2480	$5.2020	$+3.38	$1.00	$10,077.61
2025-05-03 12:00	2025-05-03 16:00	SHORT	$5.1590	$5.1280	$+2.00	$1.00	$10,079.61
2025-05-04 04:00	2025-05-04 08:00	SHORT	$5.0930	$5.0740	$+0.87	$1.00	$10,080.48
2025-05-06 08:00	2025-05-06 12:00	LONG	$4.8500	$4.8770	$+1.78	$1.00	$10,082.26
2025-05-07 04:00	2025-05-07 08:00	SHORT	$4.9620	$4.8780	$+7.46	$1.00	$10,089.73
2025-05-08 12:00	2025-05-09 00:00	LONG	$5.5460	$6.1780	$+55.98	$1.00	$10,145.70
2025-05-09 12:00	2025-05-09 16:00	LONG	$6.2590	$6.2990	$+2.20	$1.00	$10,147.90
2025-05-09 20:00	2025-05-10 04:00	SHORT	$6.3530	$6.6040	$-20.75	$1.00	$10,127.15
2025-05-10 08:00	2025-05-10 12:00	SHORT	$6.5440	$6.7530	$-16.97	$1.00	$10,110.18

2. Resultados -- C-4h

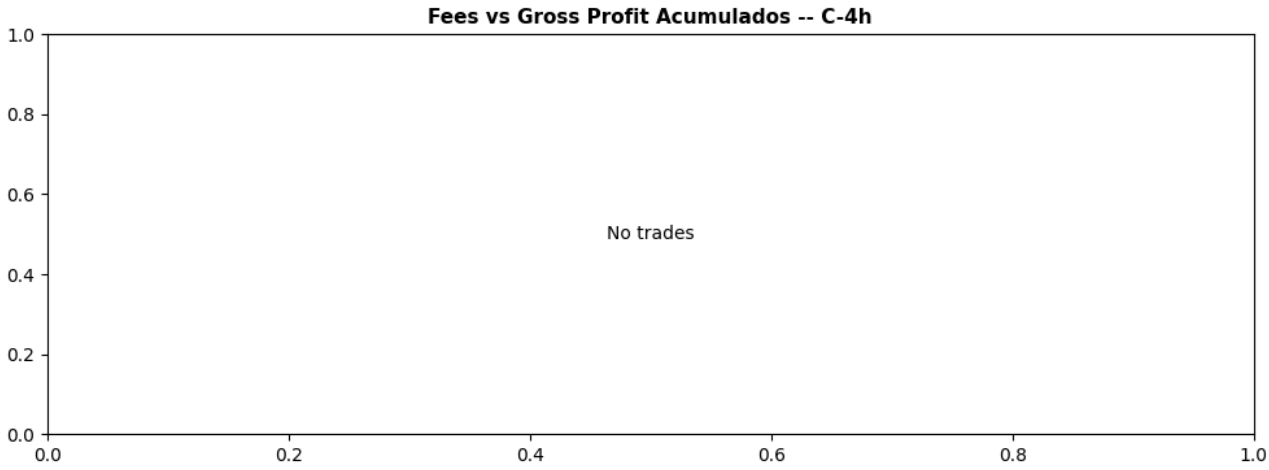
2.1 Tabla de P&L en USD

```
Capital inicial:        $   10,000.00
Capital final:          $   10,000.00
Profit/Loss neto:       $       +0.00  (+0.00%)
---------------------------------------------
Total trades:                        0
Trades LONG:                         0
Trades SHORT:                        0
Periodos en FLAT:           2184  (100.0%)
---------------------------------------------
Trades ganadores:               0  (0.0%)
Trades perdedores:                   0
---------------------------------------------
Gross profit:           $        0.00
Total fees pagadas:     $        0.00
Net profit:             $       +0.00
---------------------------------------------
Mejor trade:            $       +0.00
Peor trade:             $       +0.00
Racha ganadora max:            0 trades
Racha perdedora max:           0 trades
---------------------------------------------
Sharpe ratio:                   0.000
Max drawdown:               0.00%  ($0.00)
Dias en test:                    2184
```

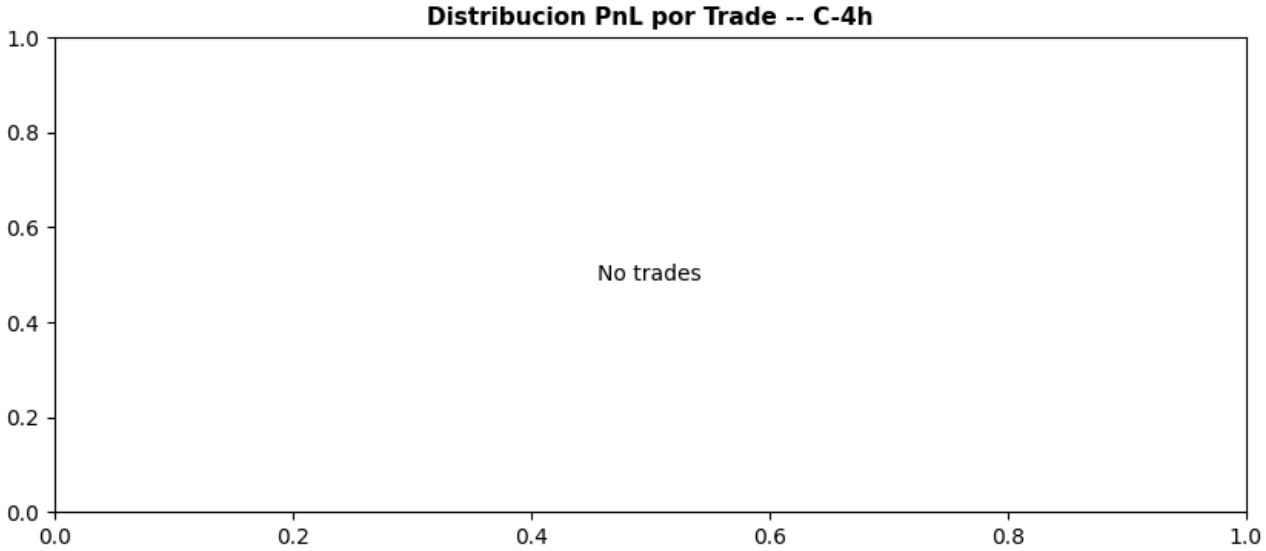
2.2 Equity Curve en USD



2.3 Fees Acumuladas vs Profit Bruto



2.4 Distribucion PnL por Trade



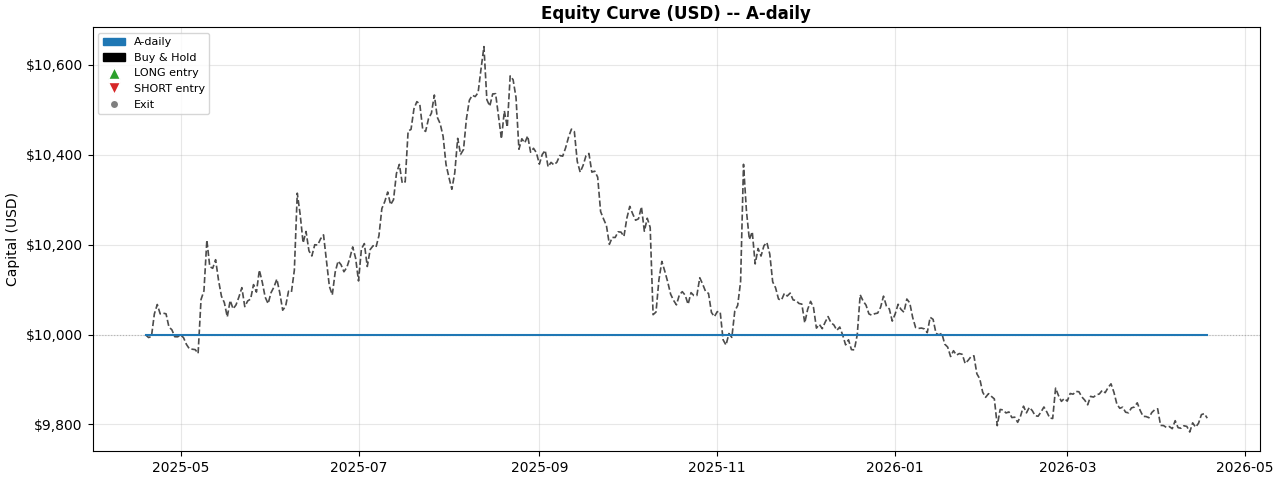
NOTA: Esta combinacion produjo 0 trades durante el periodo de test.
El modelo no genero probabilidades que crucen los umbrales configurados.
Resultado: capital permanecio en $10,000 (100% FLAT).
Ver seccion de errores para detalles de la distribucion de probabilidades.

2. Resultados -- A-daily

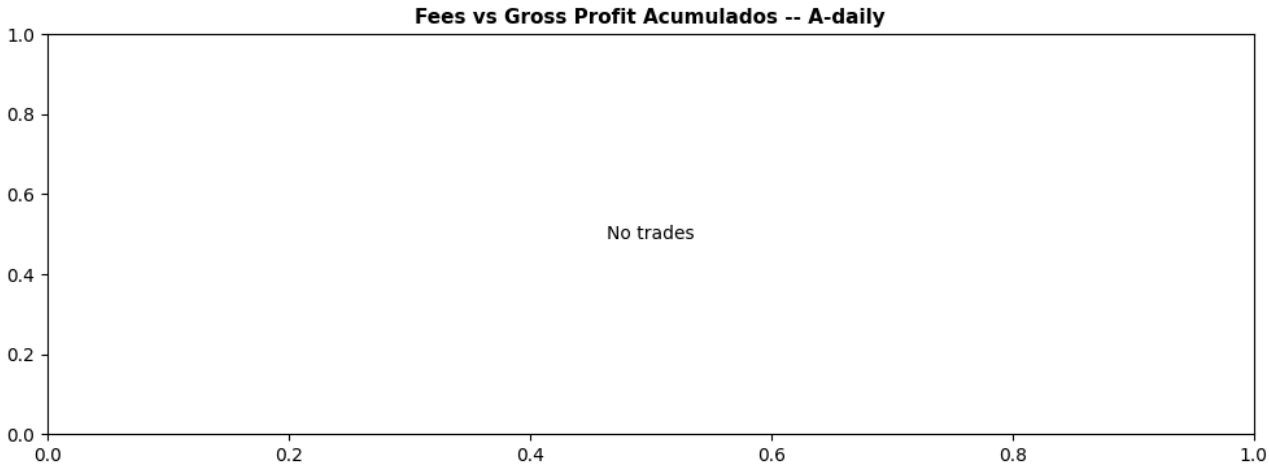
2.1 Tabla de P&L en USD

```
Capital inicial:        $   10,000.00
Capital final:          $   10,000.00
Profit/Loss neto:       $       +0.00  (+0.00%)
--------------------------------------------
Total trades:                        0
Trades LONG:                         0
Trades SHORT:                        0
Periodos en FLAT:            365  (100.0%)
--------------------------------------------
Trades ganadores:              0  (0.0%)
Trades perdedores:                   0
--------------------------------------------
Gross profit:           $        0.00
Total fees pagadas:     $        0.00
Net profit:             $       +0.00
--------------------------------------------
Mejor trade:            $       +0.00
Peor trade:             $       +0.00
Racha ganadora max:            0 trades
Racha perdedora max:           0 trades
--------------------------------------------
Sharpe ratio:                   0.000
Max drawdown:               0.00%  ($0.00)
Dias en test:                     365
```

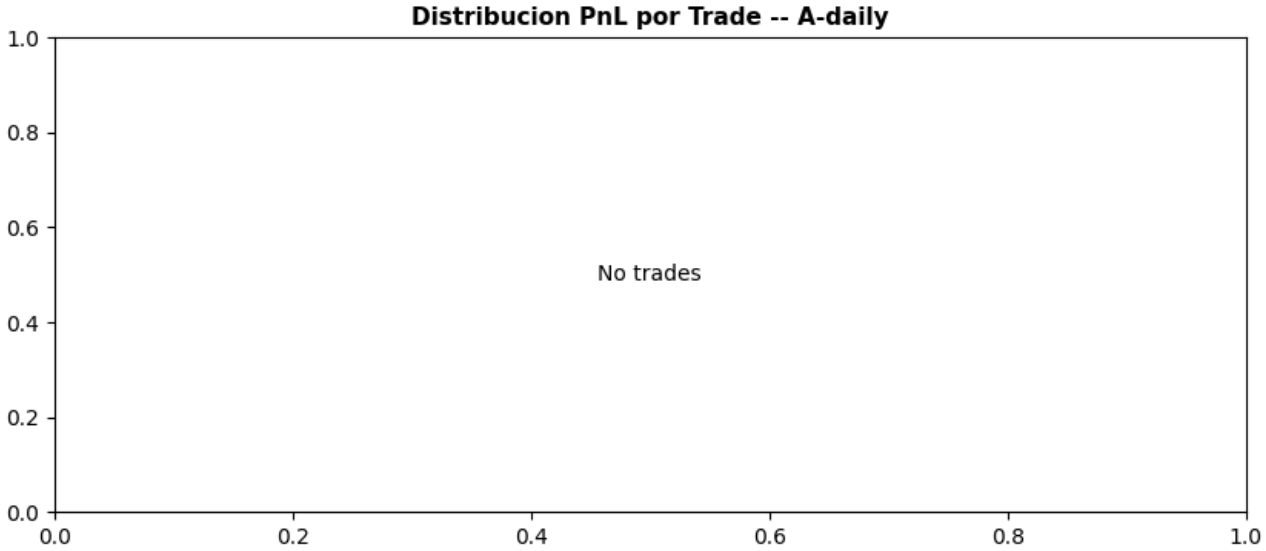
2.2 Equity Curve en USD



2.3 Fees Acumuladas vs Profit Bruto



2.4 Distribucion PnL por Trade



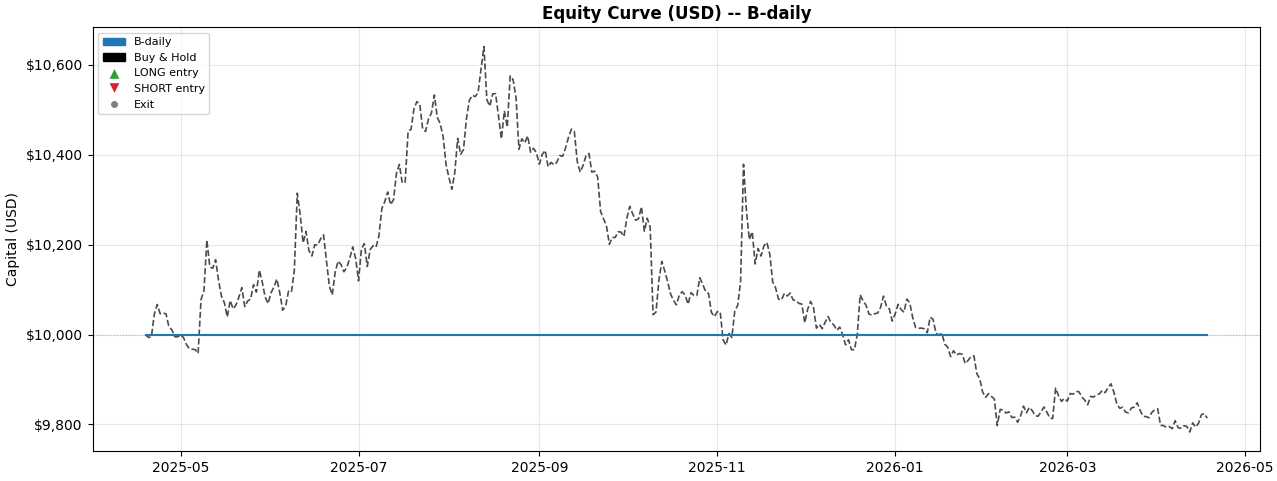
NOTA: Esta combinacion produjo 0 trades durante el periodo de test.
El modelo no genero probabilidades que crucen los umbrales configurados.
Resultado: capital permanecio en $10,000 (100% FLAT).
Ver seccion de errores para detalles de la distribucion de probabilidades.

2. Resultados -- B-daily

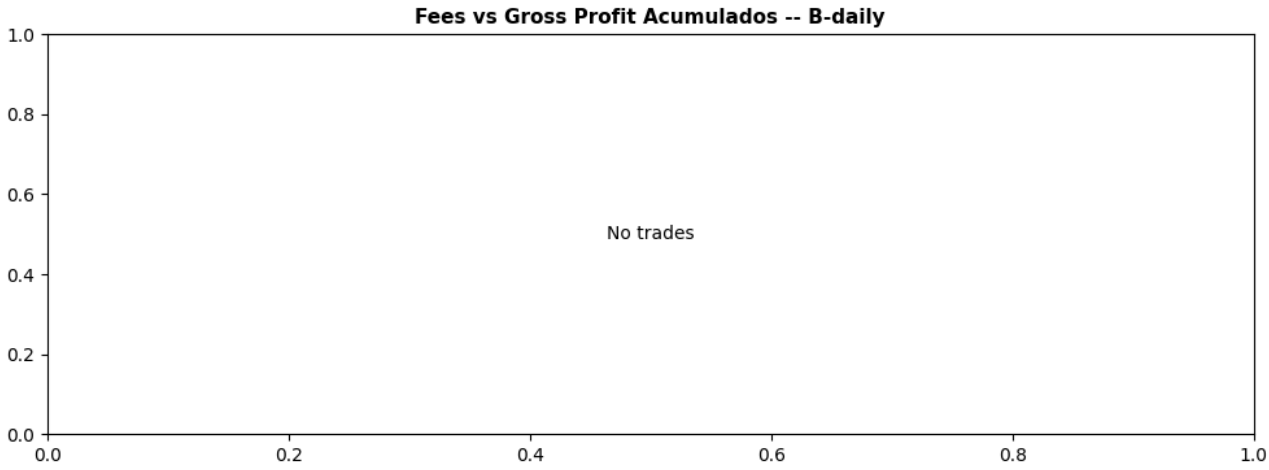
2.1 Tabla de P&L en USD

```
Capital inicial:        $   10,000.00
Capital final:          $   10,000.00
Profit/Loss neto:       $       +0.00  (+0.00%)
---------------------------------------------
Total trades:                        0
Trades LONG:                         0
Trades SHORT:                        0
Periodos en FLAT:              365  (100.0%)
---------------------------------------------
Trades ganadores:                0  (0.0%)
Trades perdedores:                   0
---------------------------------------------
Gross profit:           $        0.00
Total fees pagadas:     $        0.00
Net profit:             $       +0.00
---------------------------------------------
Mejor trade:            $       +0.00
Peor trade:             $       +0.00
Racha ganadora max:             0 trades
Racha perdedora max:            0 trades
---------------------------------------------
Sharpe ratio:                   0.000
Max drawdown:                0.00%  ($0.00)
Dias en test:                      365
```

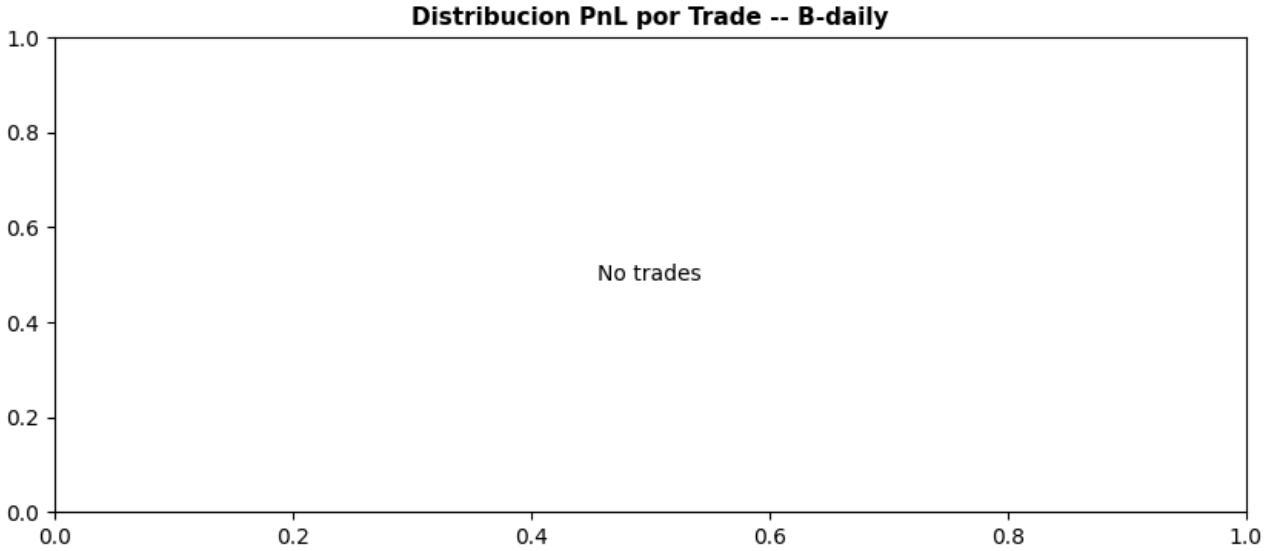
2.2 Equity Curve en USD



2.3 Fees Acumuladas vs Profit Bruto



2.4 Distribucion PnL por Trade



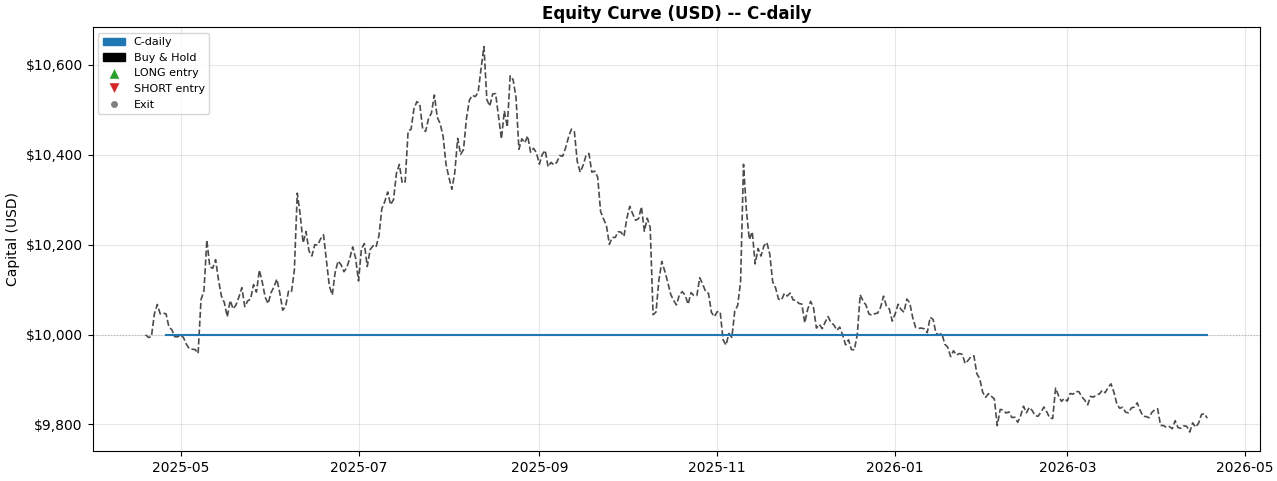
NOTA: Esta combinacion produjo 0 trades durante el periodo de test.
El modelo no genero probabilidades que crucen los umbrales configurados.
Resultado: capital permanecio en $10,000 (100% FLAT).
Ver seccion de errores para detalles de la distribucion de probabilidades.

2. Resultados -- C-daily

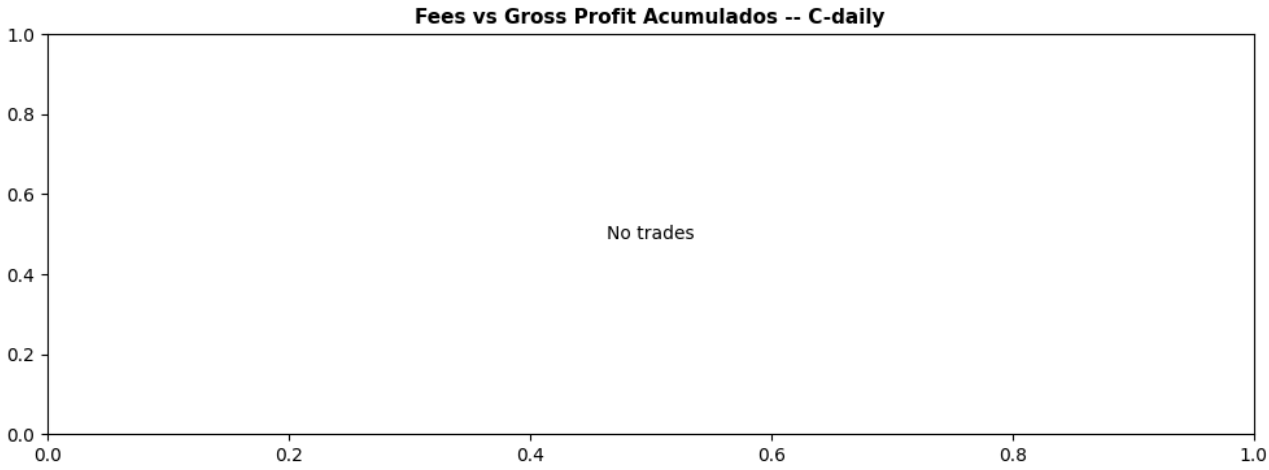
2.1 Tabla de P&L en USD

```
Capital inicial:        $   10,000.00
Capital final:          $   10,000.00
Profit/Loss neto:       $       +0.00  (+0.00%)
---------------------------------------------
Total trades:                        0
Trades LONG:                         0
Trades SHORT:                        0
Periodos en FLAT:            358  (100.0%)
---------------------------------------------
Trades ganadores:              0  (0.0%)
Trades perdedores:                   0
---------------------------------------------
Gross profit:           $        0.00
Total fees pagadas:     $        0.00
Net profit:             $       +0.00
---------------------------------------------
Mejor trade:            $       +0.00
Peor trade:             $       +0.00
Racha ganadora max:            0 trades
Racha perdedora max:           0 trades
---------------------------------------------
Sharpe ratio:                   0.000
Max drawdown:               0.00%  ($0.00)
Dias en test:                     358
```

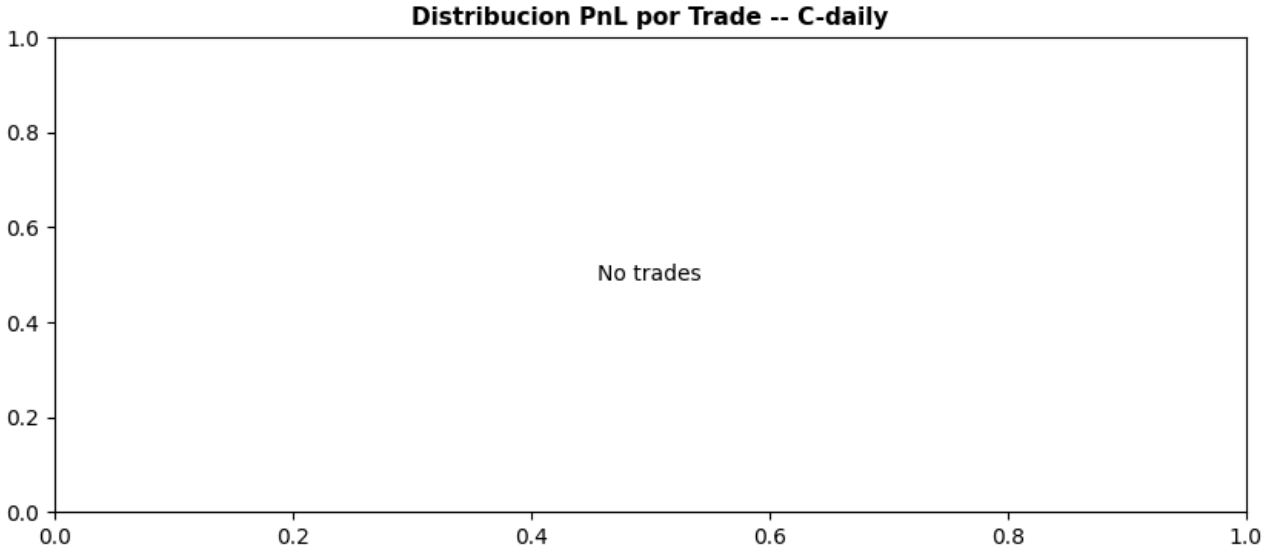
2.2 Equity Curve en USD



2.3 Fees Acumuladas vs Profit Bruto



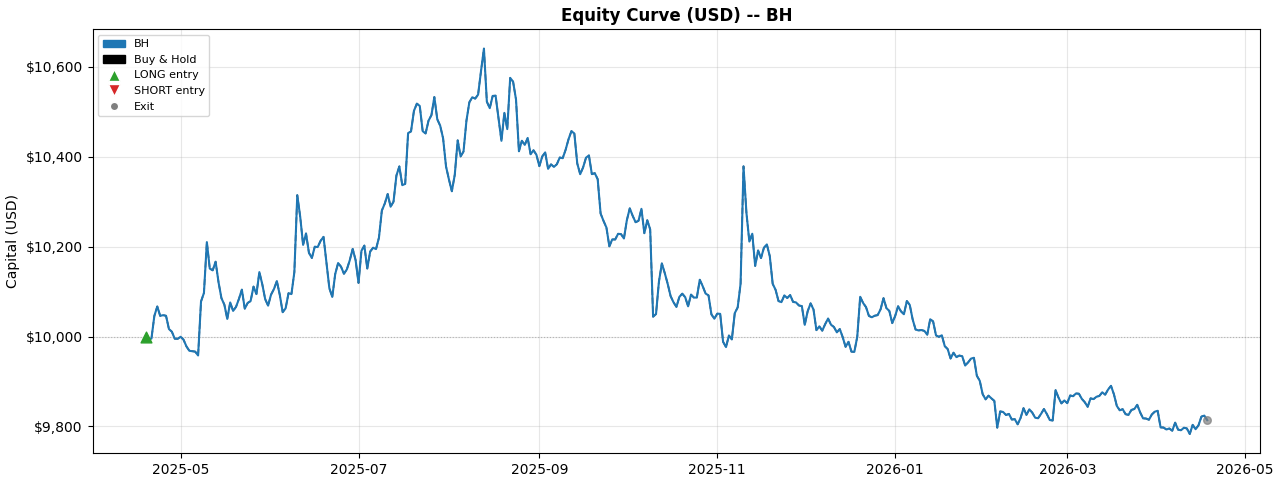
2.4 Distribucion PnL por Trade



NOTA: Esta combinacion produjo 0 trades durante el periodo de test.
El modelo no genero probabilidades que crucen los umbrales configurados.
Resultado: capital permanecio en $10,000 (100% FLAT).
Ver seccion de errores para detalles de la distribucion de probabilidades.

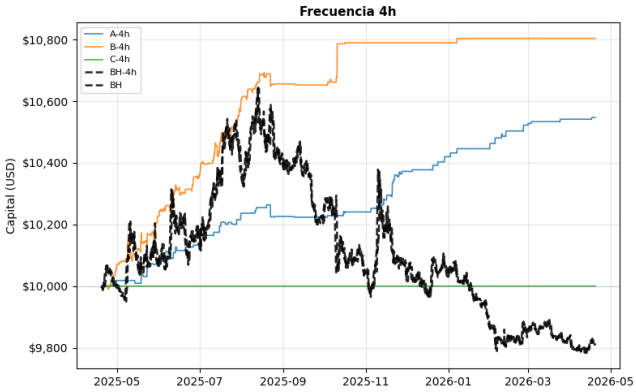
3. Benchmark -- Buy & Hold

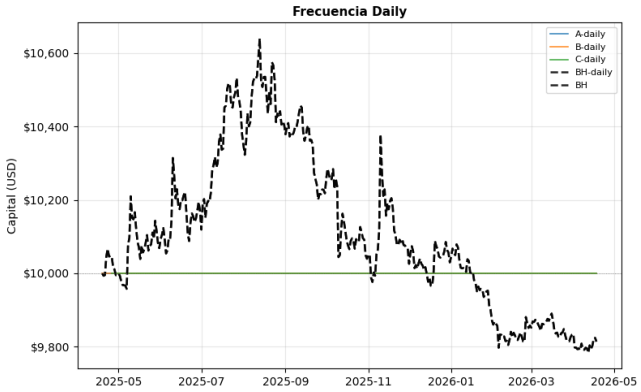
```
Capital inicial:        $   10,000.00
Capital final:          $    9,814.03
Profit/Loss neto:       $     -185.97  (-1.86%)
---------------------------------------------
Total trades:                       1
Trades LONG:                        1
Trades SHORT:                       0
Periodos en FLAT:              0  (0.0%)
---------------------------------------------
Trades ganadores:              0  (0.0%)
Trades perdedores:                  1
---------------------------------------------
Gross profit:           $        0.00
Total fees pagadas:     $        1.00
Net profit:             $     -185.97
---------------------------------------------
Mejor trade:            $     -185.97
Peor trade:             $     -185.97
Racha ganadora max:            0 trades
Racha perdedora max:           1 trades
---------------------------------------------
Sharpe ratio:                  -0.203
Max drawdown:              -8.05%  ($-856.87)
Dias en test:                     364
```



4. Tabla Maestra Comparativa

Var	Freq	Trades	%Flat	Fees	Net P&L	Ret%	Sharpe	MaxDD
A	4h	77	96.1%	$77.00	$+547.09	+5.47%	3.889	-0.40%
B	4h	167	89.5%	$167.00	$+804.08	+8.04%	3.407	-0.66%
C	4h	0	100.0%	$0.00	$+0.00	+0.00%	0.000	0.00%
A	daily	0	100.0%	$0.00	$+0.00	+0.00%	0.000	0.00%
B	daily	0	100.0%	$0.00	$+0.00	+0.00%	0.000	0.00%
C	daily	0	100.0%	$0.00	$+0.00	+0.00%	0.000	0.00%
BH	-	1	0.0%	$1.00	$-185.97	-1.86%	-0.203	-8.05%



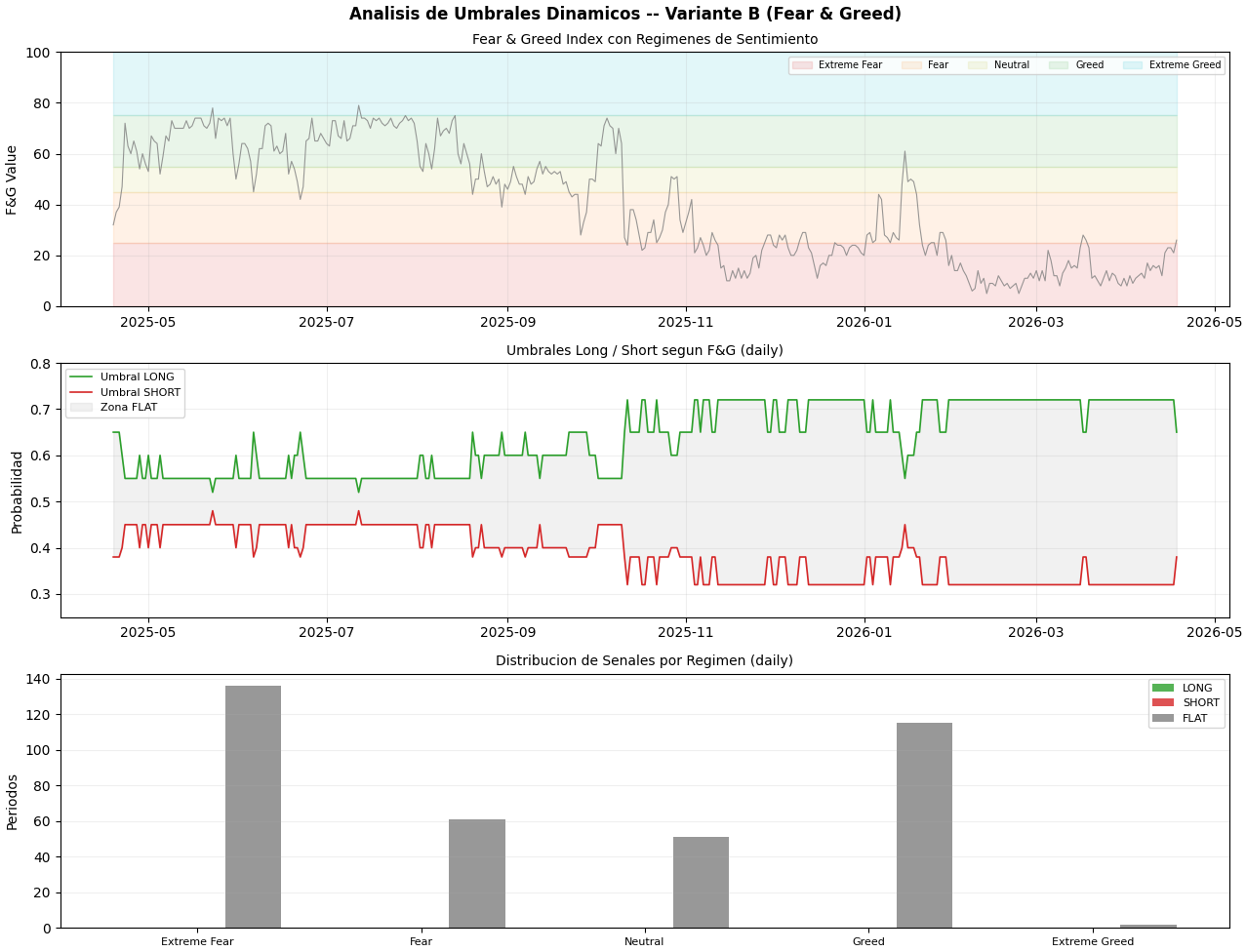


5. Analisis de Umbrales Dinamicos -- Variante B

La variante B ajusta los umbrales de entrada/salida segun el Fear & Greed Index:

- En 'Extreme Fear': exige mas conviccion para LONG (0.72) y menos para SHORT (0.32).
  Logica: mercado cae mas facil en panico, vale mas apostar al short.
- En 'Extreme Greed': umbrales casi simetricos (0.52/0.48), zona FLAT muy estrecha.
  Logica: en euforia las senales son mas confiables en ambas direcciones.

Nota: el LSTM (variante C) NO usa este mecanismo porque Fear & Greed ya es un feature de entrenamiento del modelo.



5.1 Dias por Regimen de Sentimiento (daily)

Regimen	Dias	%	LONG	SHORT	FLAT
Extreme Fear	136	37.3%	0	0	136
Fear	61	16.7%	0	0	61
Neutral	51	14.0%	0	0	51
Greed	115	31.5%	0	0	115
Extreme Greed	2	0.5%	0	0	2

6. Conclusion

Resumen de la corrida V3:

Mejor combinacion variante x frecuencia (por net profit en USD):
-> B-4h
Net P&L: $+804.08 (+8.04%)
Sharpe: 3.407
Max DD: -0.66%
Trades: 167 (89.5% en FLAT)

Observaciones generales:
* El estado FLAT reduce el numero de trades y las fees acumuladas, pero puede perder oportunidades en mercados con tendencia clara.
* La variante B (umbrales dinamicos F&G) agrega sensibilidad al sentimiento del mercado sin depender del modelo directamente.
* La variante C (LSTM) ya incorpora Fear & Greed como feature, por lo que usar umbrales fijos evita doble conteo del sentimiento.
* El trade size fijo de $500 (5% del capital) limita la exposicion maxima y evita el impacto de posiciones de gran tamano.

Disclaimer: Este reporte es solo con fines educativos. No constituye asesoramiento de inversion. Trading de criptomonedas implica riesgo de perdida total del capital invertido.

7. Errores durante la Ejecucion

* A-daily: 0 trades. El modelo XGB no cruza los umbrales. Probas: mean=0.4909 std=0.0203 max=0.5235 min=0.4507. Todos los valores caen en la zona FLAT.

* B-daily: 0 trades. El modelo XGB no cruza los umbrales. Probas: mean=0.4909 std=0.0203 max=0.5235 min=0.4507. Todos los valores caen en la zona FLAT.

* C-4h: 0 trades. El LSTM produce probas muy cercanas a 0.5. Probas: mean=0.4876 std=0.0091 max=0.5141 min=0.4582. Umbrales C: 0.58/0.42. El LSTM no entrego conviccion suficiente.

* C-daily: 0 trades. El LSTM produce probas muy cercanas a 0.5. Probas: mean=0.5053 std=0.0084 max=0.5281 min=0.4883. Umbrales C: 0.58/0.42. El LSTM no entrego conviccion suficiente.